

AMENDMENT NO. 1 SEPTEMBER 1980
TO
IS : 8337-1976 SPECIFICATION FOR
PERFORMANCE REQUIREMENTS OF
COMPRESSION JOINTS OF ALUMINIUM
CONDUCTORS IN INSULATED CABLES

Alteration

(Pages 5 and 6, clauses 4.1.1 to 4.1.3.3) — Substitute the following for the existing clauses:

4.1.1 Initial Value — The initial value of joint resistance (see Note) of the (all) specimen(s) when tested in accordance with 6.2 shall not:

- a) exceed the resistance of a length of conductor equal to the length of the compression barrel (dimension L), and
- b) differ by more than 50 percent or 5 micro-ohm (whichever is greater) from the mean of the initial joint resistance values of the six specimens prepared in accordance with 5.1.

NOTE — Joint resistance = $\frac{\text{Specimen resistance AB} - \text{Free conductor resistance}}{2}$
(see Fig. 2)

4.1.2 Values Measured in the Course of Short-Circuit Test — The joint resistance of the (all) specimen(s) measured after any one of the three applications of short-circuit current in accordance with 6.3 shall not exceed the initial joint resistance values by more than 50 percent or 5 micro-ohm, whichever is greater.

4.1.3 Values Measured after Electrical Load Cycling

4.1.3.1 The average of six final values of joint resistance of the test specimen taken over 1000 cycles shall not exceed the average equivalent conductor resistance taken over the same period by more than 50 percent or 5 micro-ohms, whichever is greater.

NOTE — The equivalent conductor resistance is the resistance of the length of the control conductor of length equal to half the ferrule length (which is marked $2L$ in Fig. 2).

4.1.3.2 The average of six final values of joint resistance of the test specimen taken over 1000 load cycles shall not exceed the resistance of the test specimen as measured immediately before the start of the electrical load cycle test, by more than 50 percent or 5 micro-ohms, whichever is greater.

4.1.3.3 The graph of joint resistance against number of load cycles shall demonstrate with a reasonable probability that the change (rise or fall) of the resistance over 1 000 cycles is not more than 20 percent of the average resistance over the same period.

NOTE — If the graph of resistance is unacceptable to either purchaser or supplier, a statistical method (under consideration) shall be used to determine the acceptability of a connector or terminal end.

(ETDC 59)